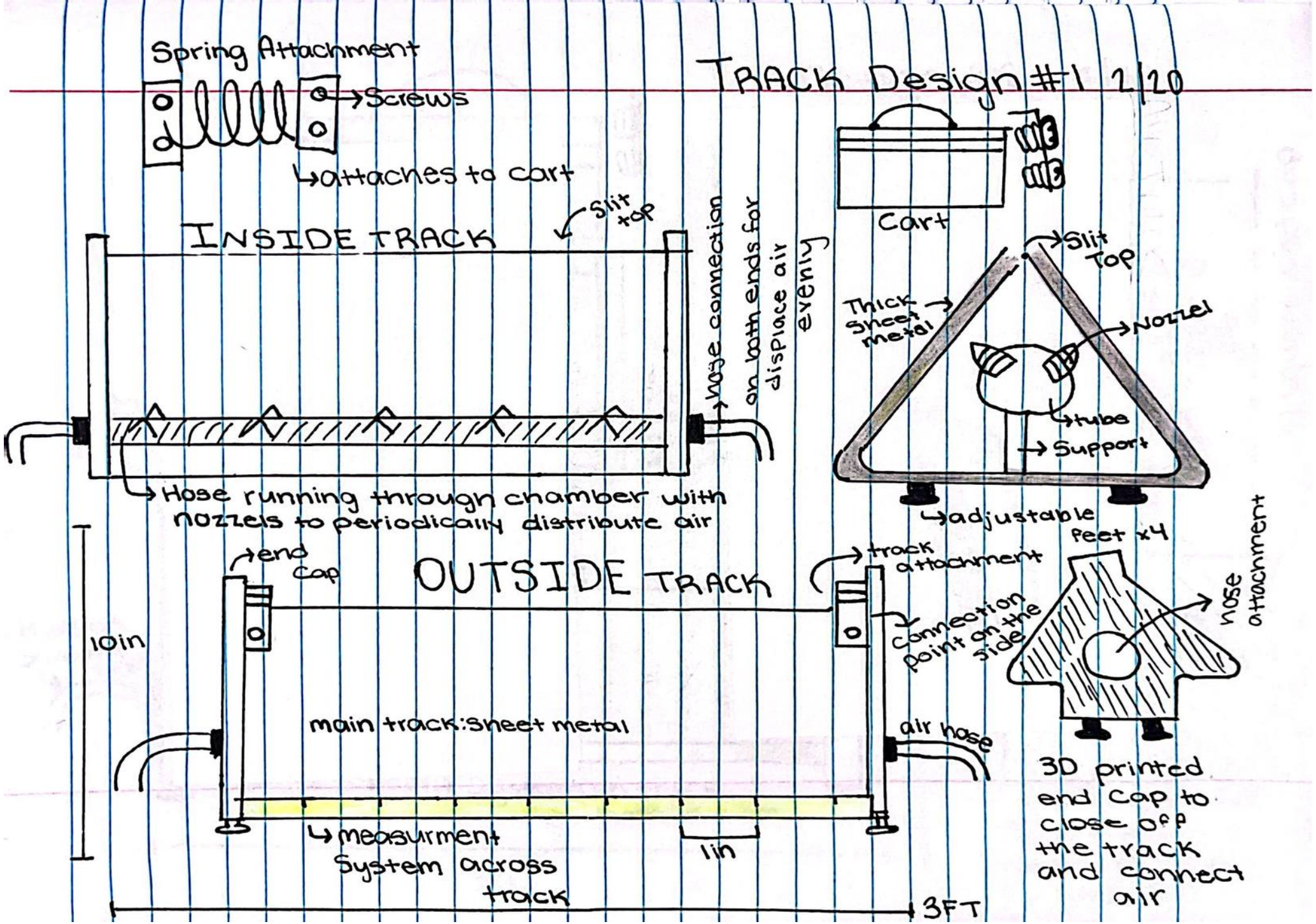
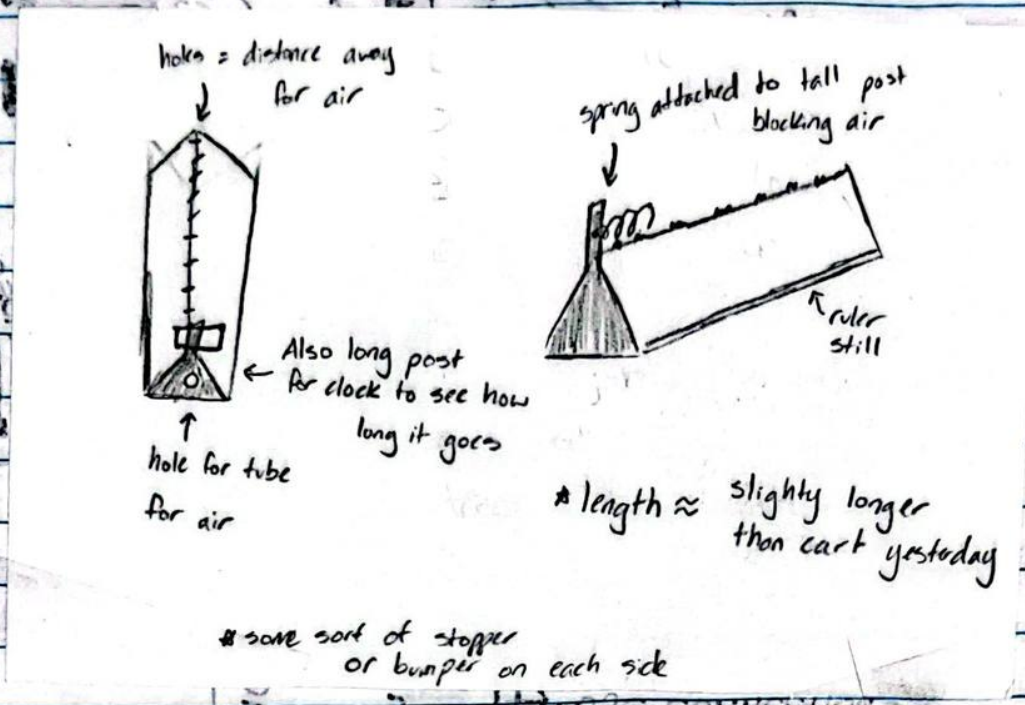
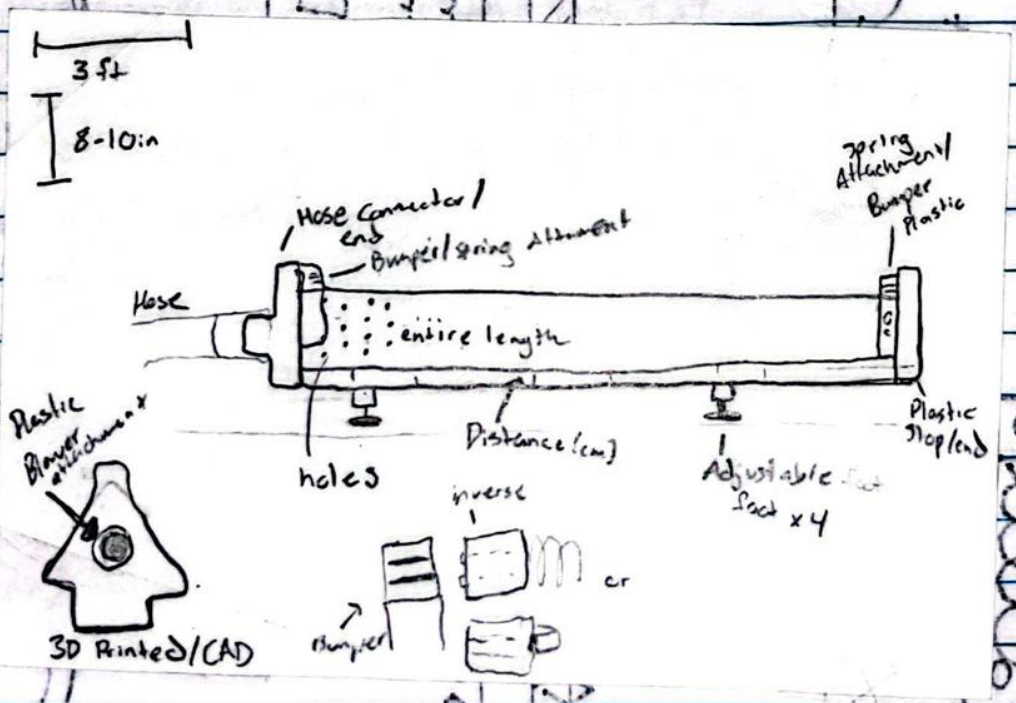
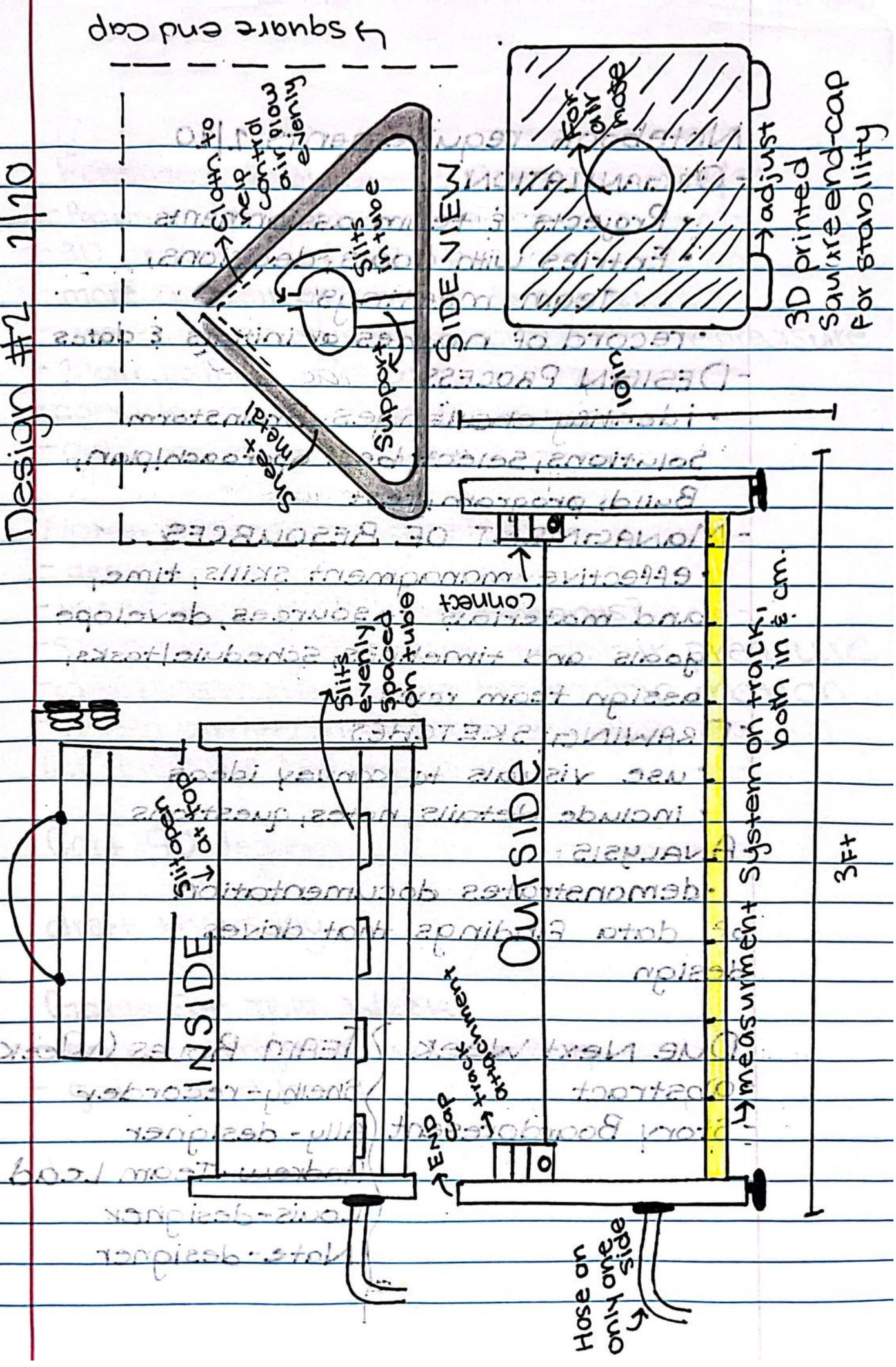
TRACK Design #1 2/20




holes = distance away for air
Also long post for clock to see how long it goes
hole for tube for air
spring attached to tall post blocking air
ruler still
* length ≈ slightly longer than cart yesterday
* some sort of stopper or bumper on each side


3 ft
8-10 in
Hose Connector / end
Bumper/Spring Attachment
Spring Attachment / Bumper Plastic
Hose
entire length
Plastic Blower attachment
holes
Distance (cm)
Adjustable
Plastic Stop/end
inverse
or
bumper
3D Printed/CAD

Design #2 1/20


square end cap
SIDE VIEW
Slits
Support
Join
3D printed square end-cap for stability
adjust
for
Slits evenly spaced on tube
Connect
INSIDE
OUTSIDE
Measurement System on track, both in cm.
3Ft
Slits to open
Track cap
Hose or wire

Feedback from Group Presentation 2/25

- focus on physics aspect
- 3D printed nozzel/pipes
- more air will escape in design #1
- use fluid analysis to calculate air preasure
- 2 air pumps was a popular vote
- consider air coming from bottom
- add more cart attachments

Notes From Tuesday 2/25

- design end cap
- interchangable attachments?
- skip tubing for now, test air preasure
- demonstrate simple harmonic motion
- tested air preasure in circle tube, air preasure felt the same.

Cart: 90 degrees

draft: thursday

Goals For This Week

- Present Project successfully
- gain Feed back

1

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Abstract design ideas (due Feb 27)

- square 3-D printed end caps
- fort compressed air line feeding to both ends of track
- single sheet of metal bended to 45° 45° 90° degree geometry
- design endcap attachment points designed the same as cart attachment points (universal)
- powered by air compressor
- variant pressure integrated into air compressor
- measurements provided on track in inches & centimeters
- 3 justable feet for level
- 3ft by 10in dimensions (36 x 10in)
-

~~[illegible]~~

- ~~[illegible]~~
- ~~[illegible]~~

→ ~~[illegible]~~

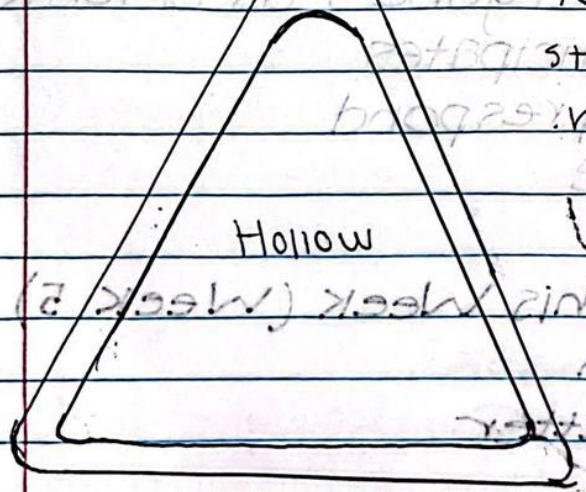
→ ~~[illegible]~~

Updated Prototype based on Poster presentation

→ start with no tubing through track

→ maybe test using fluid dynamics for air preasure

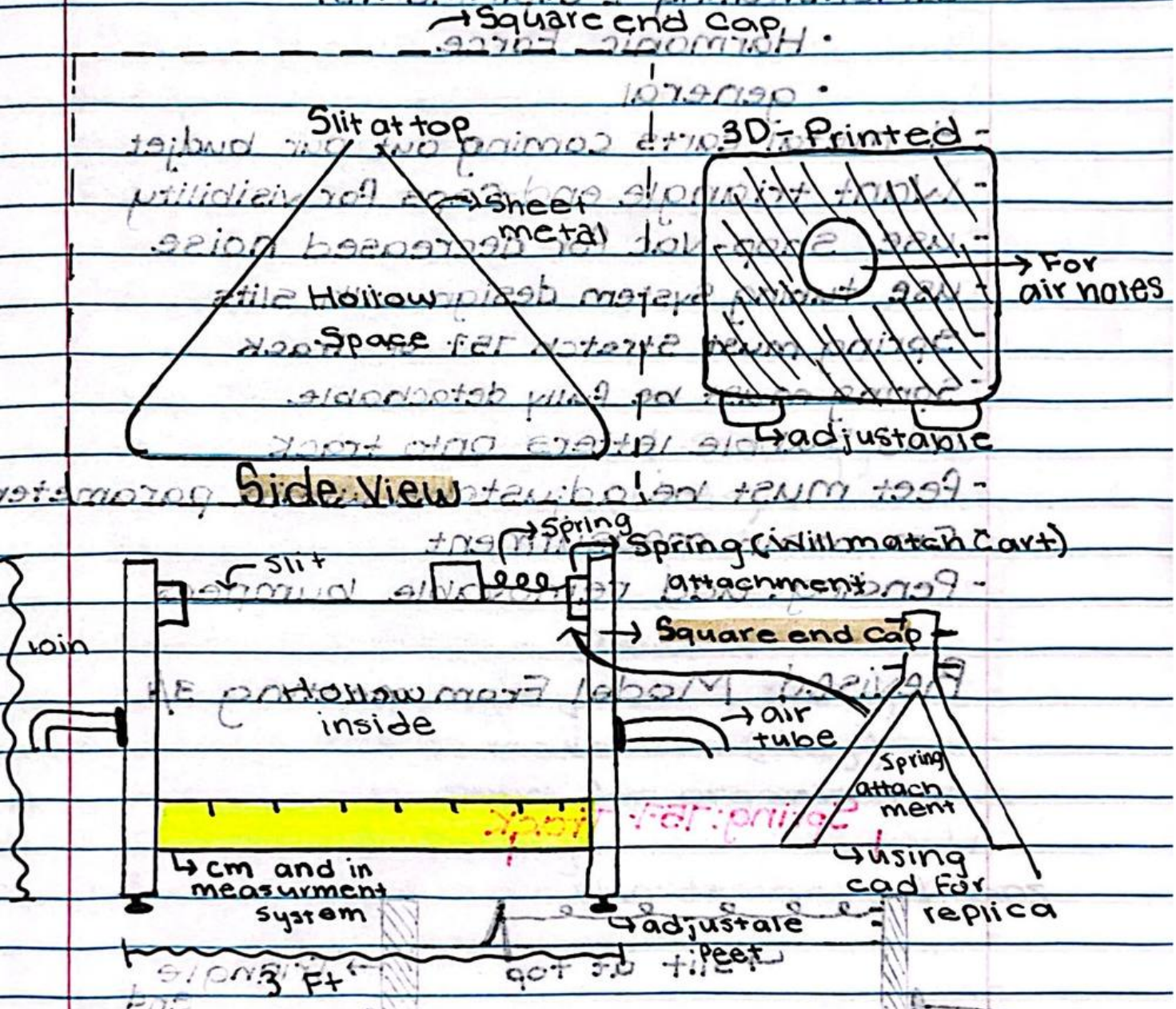
→ Confirmed: have air from both sides



New start: start with hollow inside

Inside with no tubing system

Final Revised prototype from Presentation



Prototype: Card board

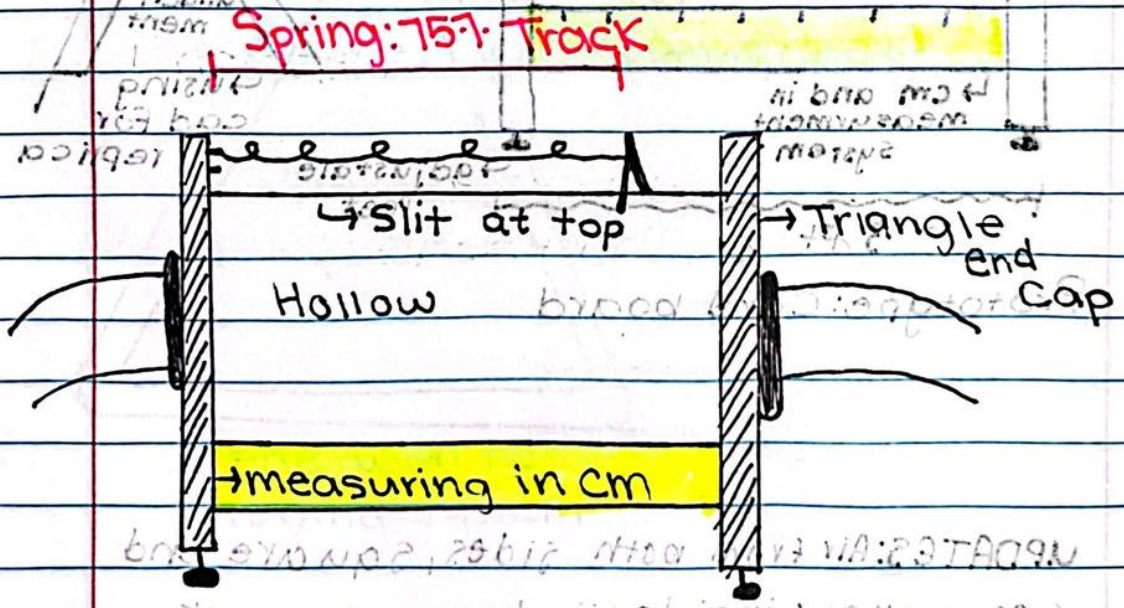
UPDATES: Air from both sides, square end cap, hollow inside (it does not work, will insert tube with nozzle (slit), use anti skid pads so it does split

meetings: Fridays @ 2:00pm

Notes From Customer meeting 2/28

- demonstrating 2 expiriments:
  - Harmonic Force
  - general
- 4 metal carts coming out our budjet
- Want triangle end-caps for visibility
- use Shop-Vac for decreased noise
- use tubing system design with slits
- Spring must stretch 75% of Track
- Spring must be fully detachable
- add bubble letters onto track
- feet must be adjustable within parameters
- Cm main measurment
- Pending: add removable bumpers

Revised Model From meeting 3/4



Changes: Triangle end Cap

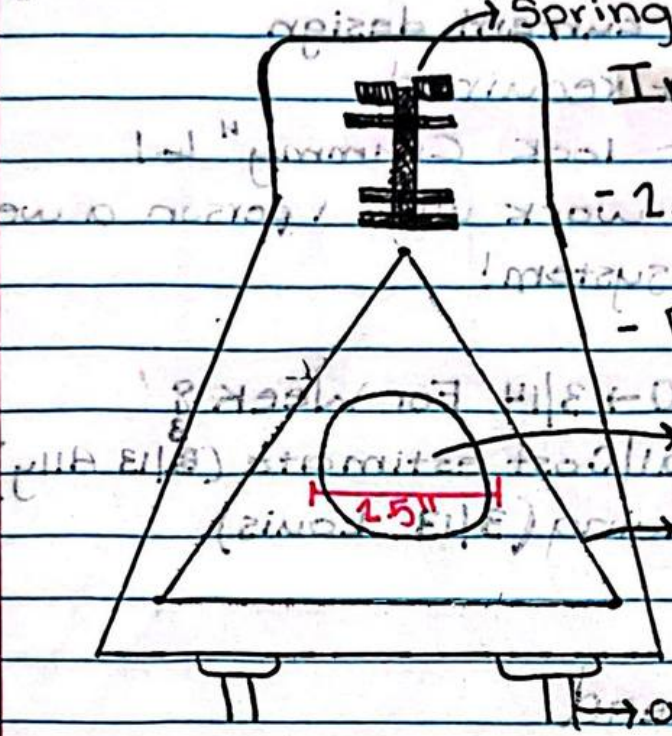
3/6 End-Cap design ideas

- 3-D print end caps
- Leave holes in print at the bottom to add adjustable legs
- have triangle inserts where track is sealed
- Have external insert for air source
- internal diameter entering track is smaller than external diameter
- Print material: PLA

New List of materials

- air: tube fan
- 1.5 inch diameter tube
- Clamps: seal air from tube fan to track
- 3D printed end caps
- track (sheet metal?)
- magnetic bubble levels
- measuring tape
- adjustable feet
- adapter to tube fan (3D print)
- springs
- cart (Provided by client, Pending)
- 10/24 inch screws

END CAP DESIGN IDEA 3/10



→ Spring attachment

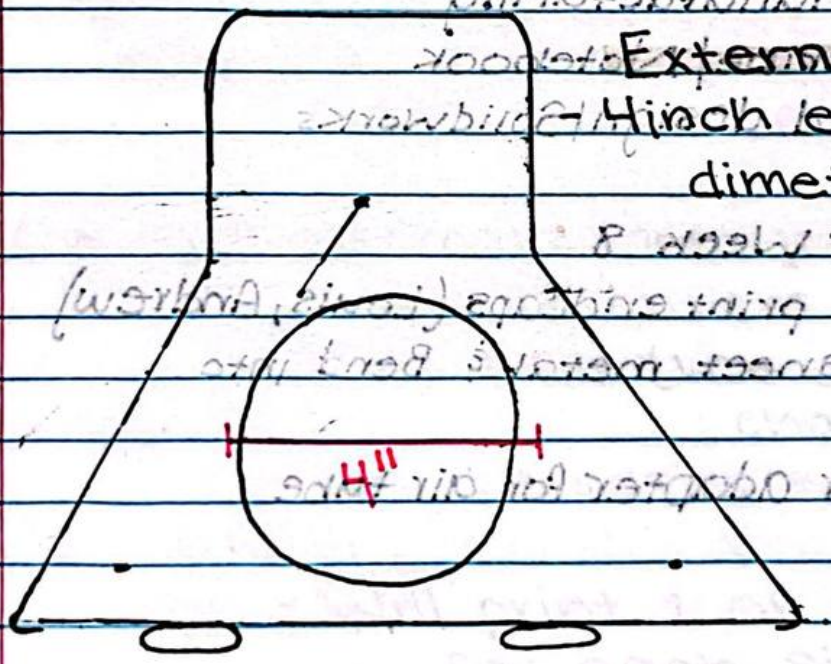
Internal View:

- 2.5in diameter air hole
- PLA Print material

→ air hole 2.5" diameter

→ inserts for track

→ adjustable feet

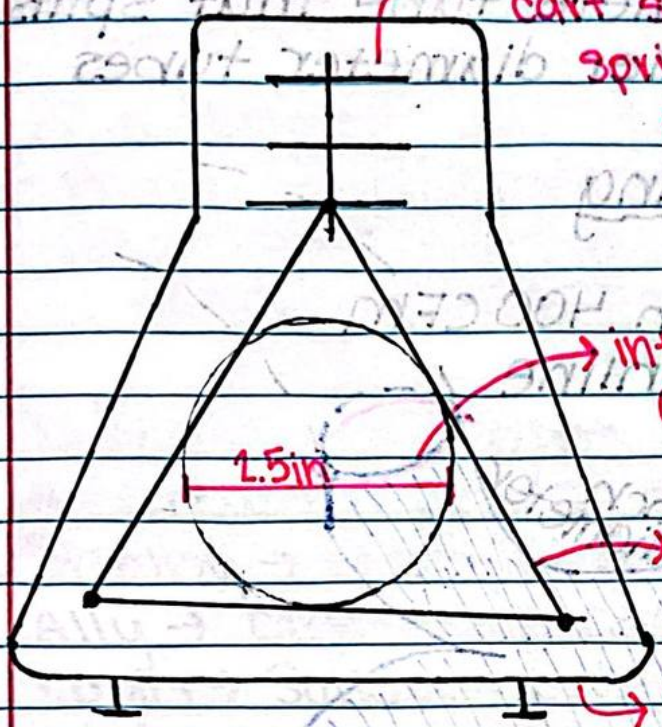


External View:

- 4inch external dimeter

- The endcaps are where the air tube from the tube fan connect to supply air to the track

New design For Endcaps



→ matches end of cart so adding/removing springs is universal and easy

→ internal and external diameter will remain the same.

→ Leave insert where we can insert caps

→ leave holes to insert adjustable feet to level track

↳ adjustable feet

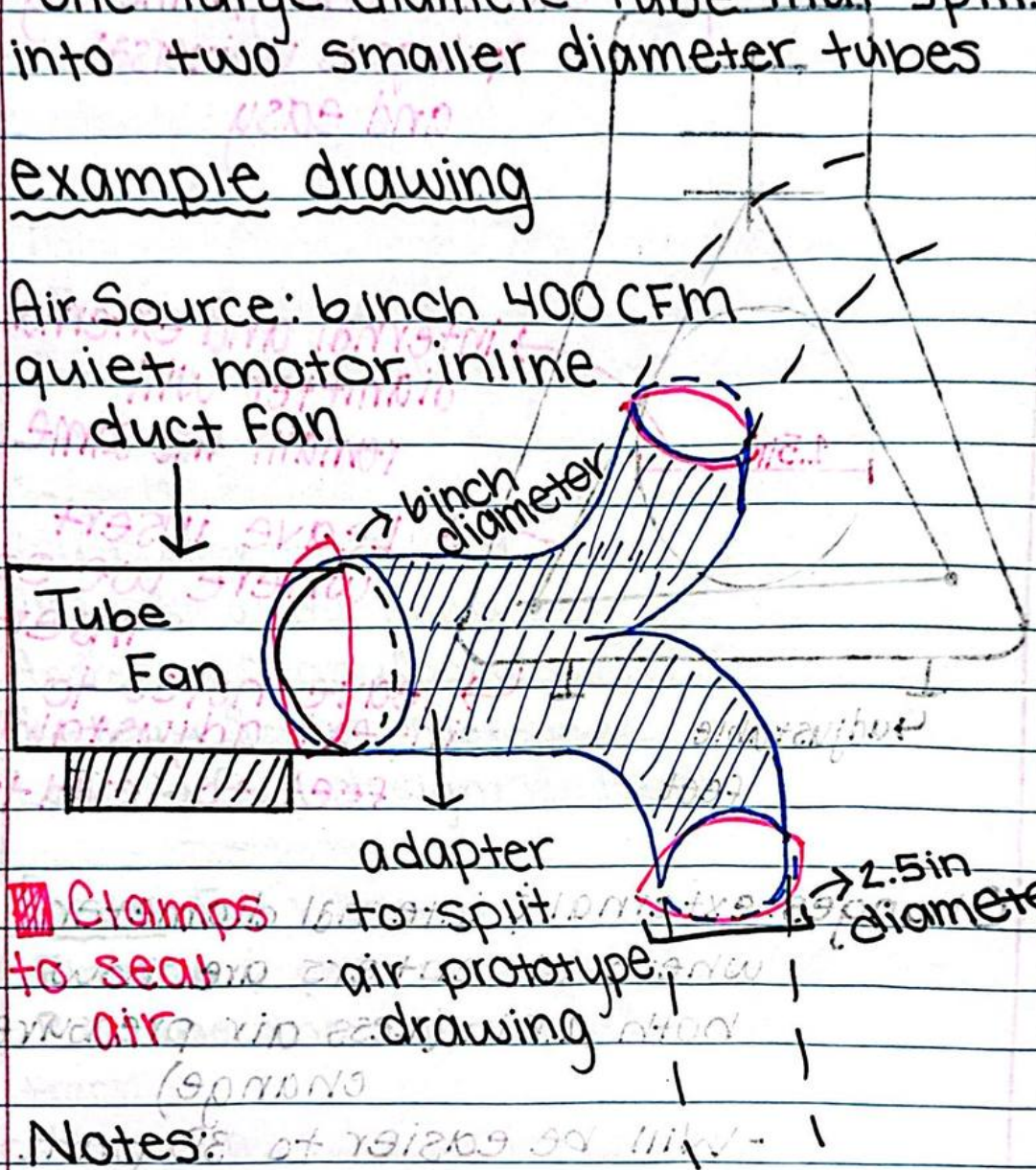
Changes: external & internal diameter where air enters are now both 2.5in (Less air preasure change)

- Will be easier to 3D print
- Will print 2 of these for each side of track

Brainstorming Air Adapter For Tube Fan

- 3D printed (PLA filiment)
- one large diamete tube that splits into two smaller diameter tubes

example drawing



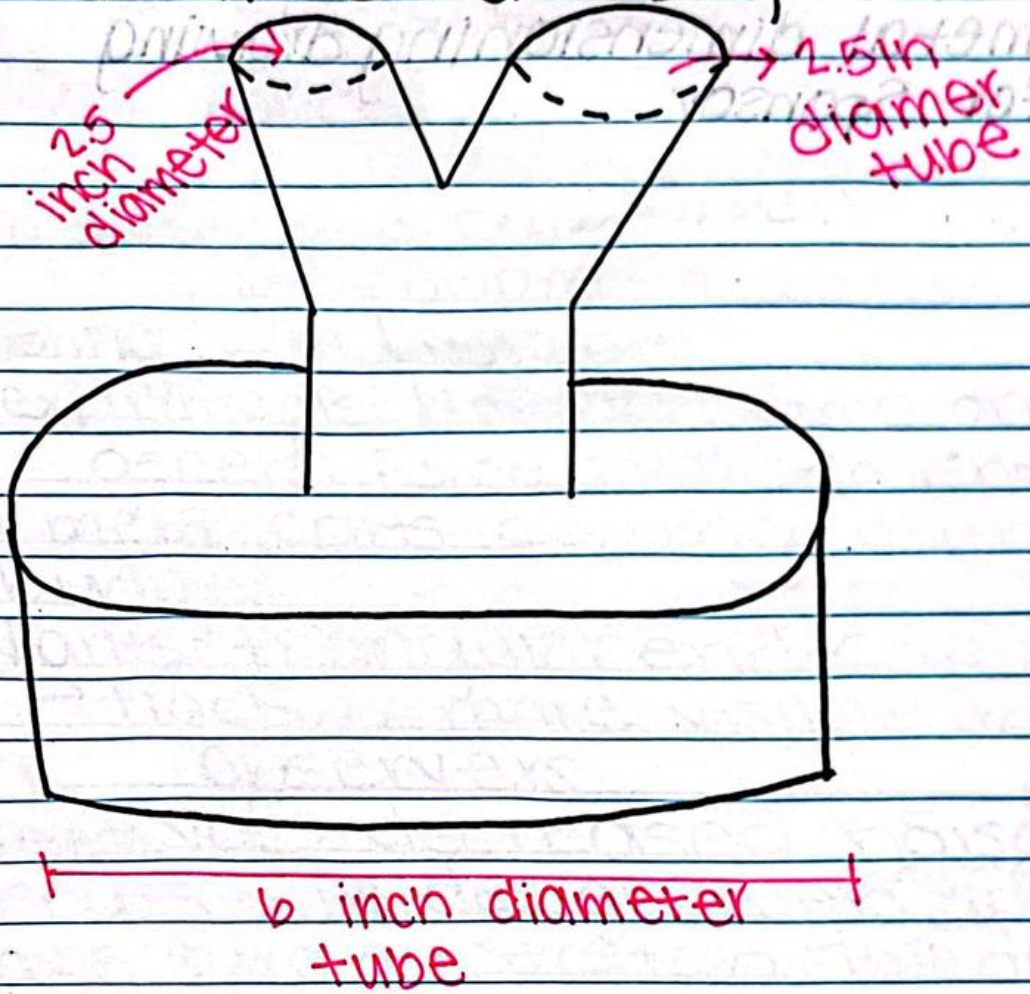
Notes:

- Need 6 inch diameter clamp from tube fan to adapter to seal air
- Need 1.5in diameter clamps to seal air from split tubes to endcaps
- Louis/Andrew CAD drawing

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OFFICIAL AIR ADAPTER DESIGN

- 3D printed (PLA filiment)
- 6 inch diameter, split into 2.5in diameter
- use 6in diameter clamp to seal air (Found on amazon)

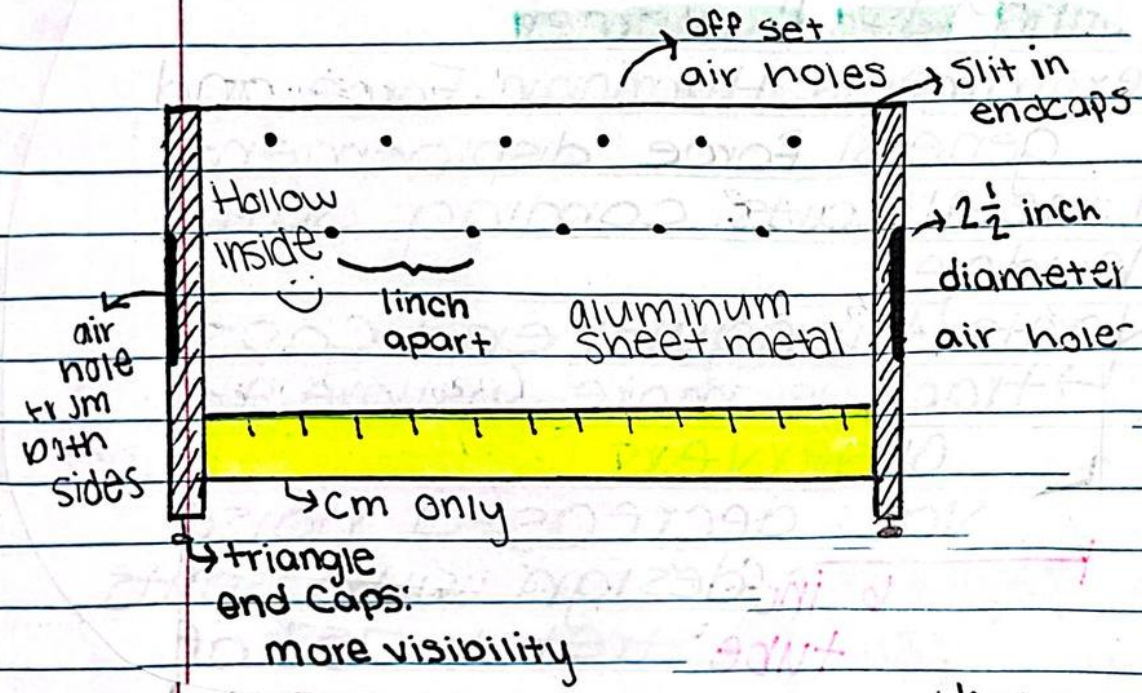


- 3D printing next week (Hopefully)
- Ordering 6in diameter Clamp (Shelby)

Week 11 ([redacted]) (3/31 → 4/4)

- 3D printed adaptor has been printed
- endcap prototype #1 has been printed
- Andrew is beginning sheet metal dimensioning drawing for Sponsor

Revised model



dimensions → 3ft x 10in

- Changes: Holes instead of slit at top
- Connected to endcap: universal spring attachment

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Adapter and clamp (4/1)

Goals For Week 11

- finalize endcap design, fix print (the 3D printer printed it weird)
- Test air splitter with tube fan in shop, feel air preasure
- Send Recia metals sheet metal dimensioning
- CDR Presentation (4/3)

Week team Goals

- discuss with team dynamic, make sure everyone is happy

Due next Week

- Updated Gantt chart (Shelby)
- Design description (Nate, Ally, Shelby, Andrew, Louis)

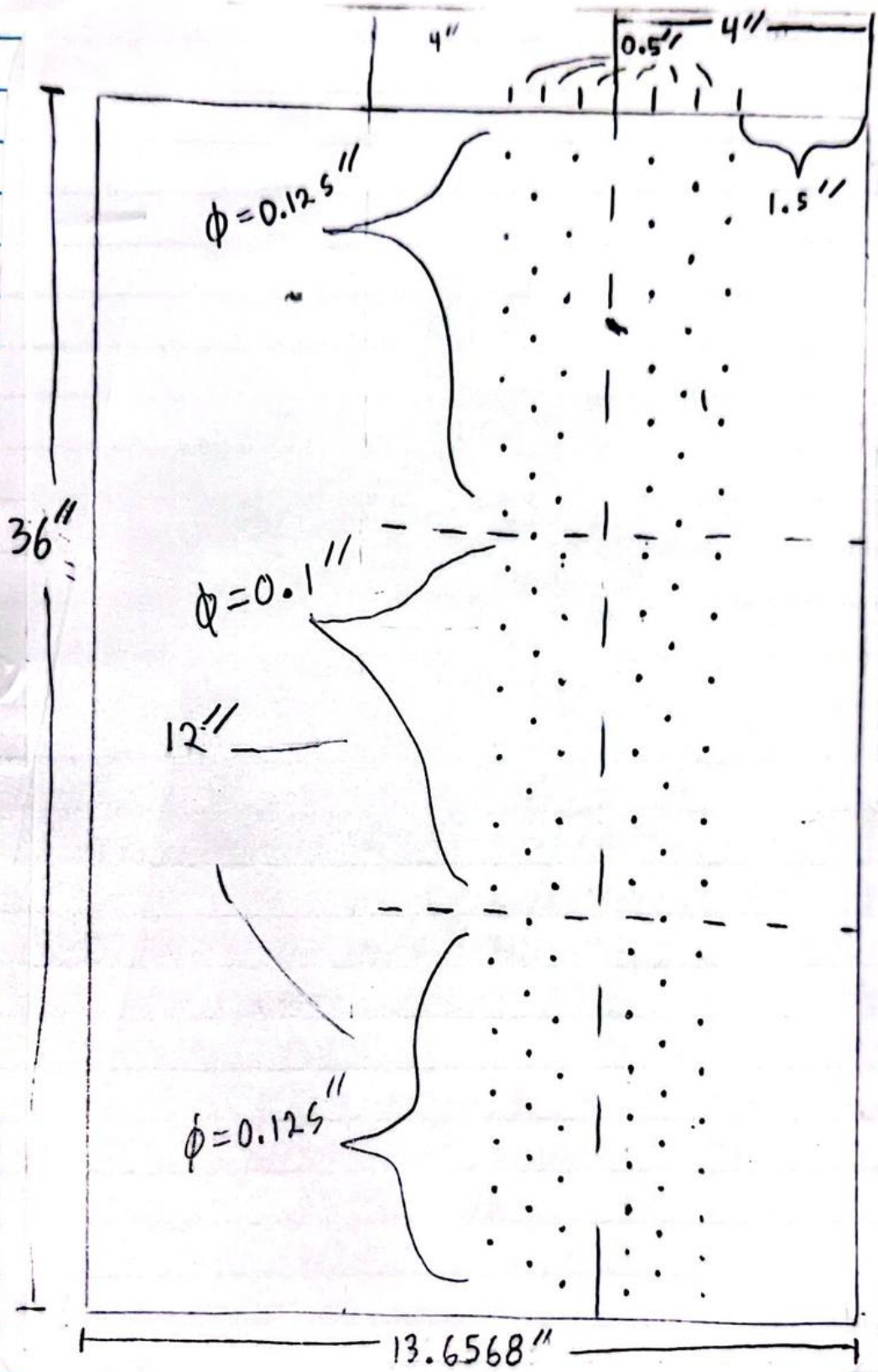
Notes From Client (From last Week)

- Will buy "gliders" for us in trade of us buying air source

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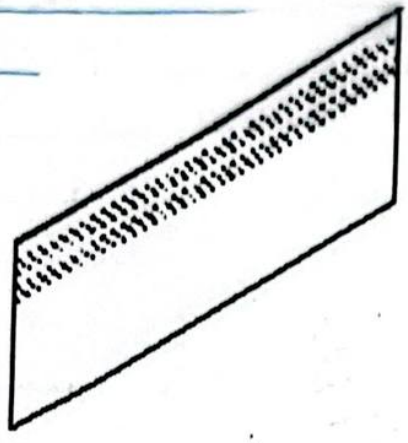
(Andrew Sent these Over)

Recla Metals: Dimensional Drawing



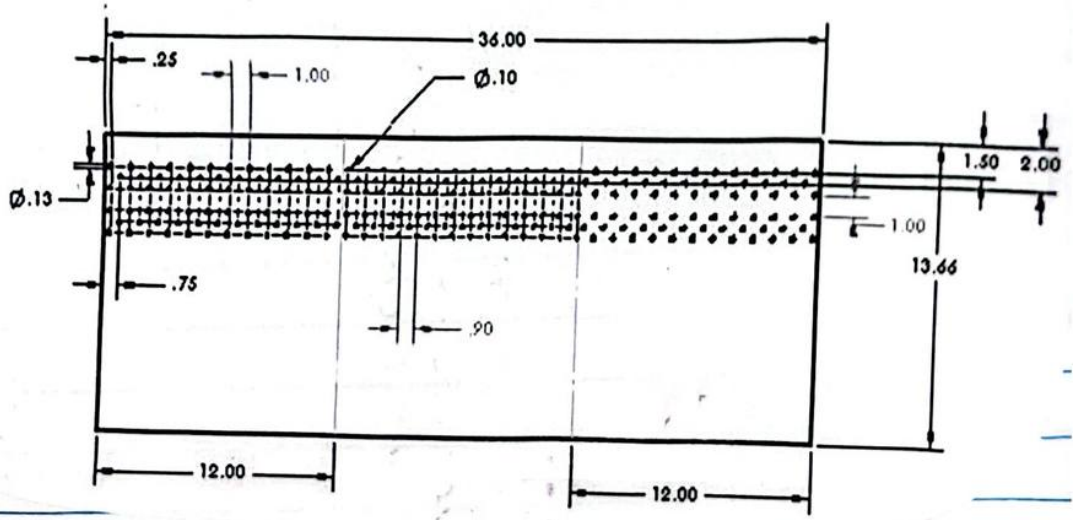
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Sheet metal dimensioning (2)



Sheet Metal Dimensioning

- The first and last 12 inches have hole diameters of .125in (1/8) and horizontal spacing of 1in and there are 12 holes per section in each row
- The middle 12 inches has hole diameters of .1in and horizontal spacing of 0.9in. There are 14 holes in the rows 1,3,4,6 and 13 holes in rows 2,5.

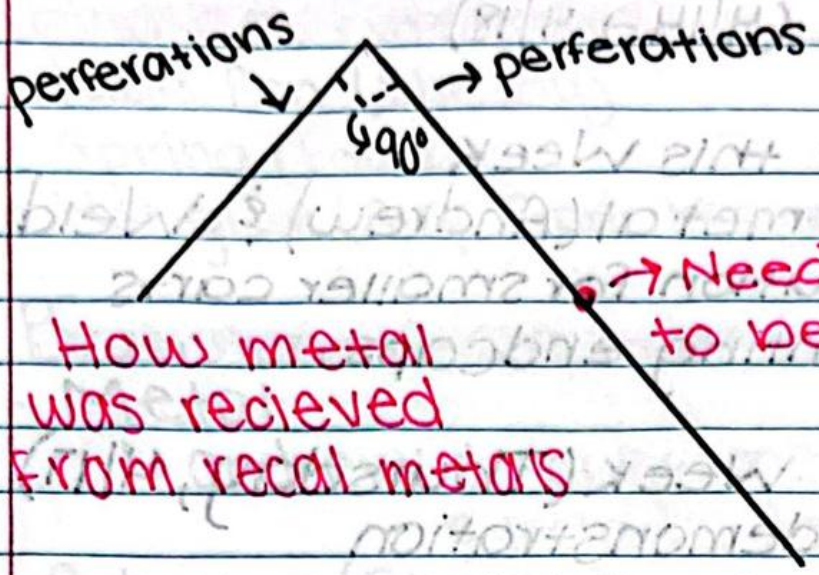


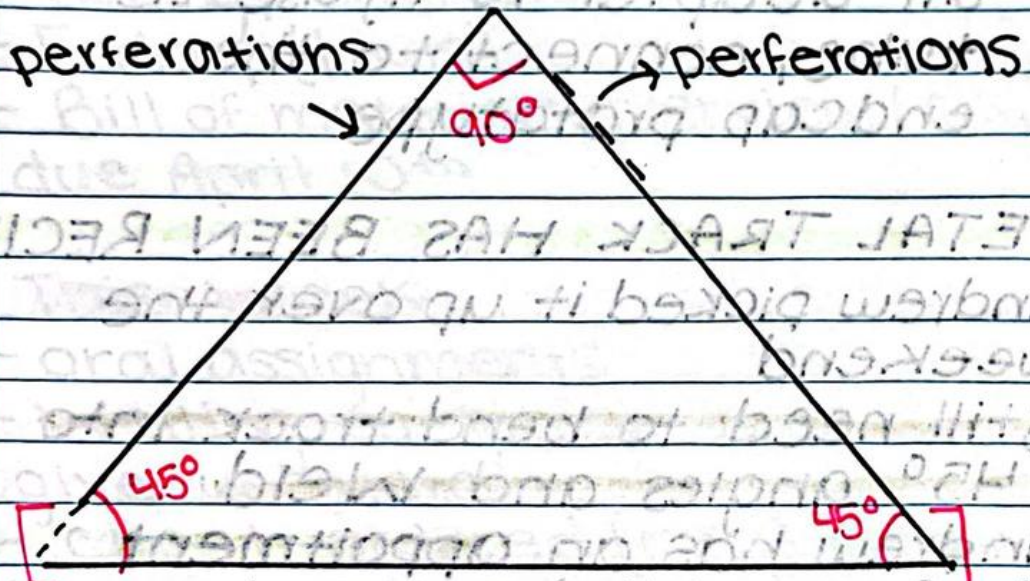
- Recla metals has confirmed these drawings...
- Will Hopefully be assembled in the next two weeks

Week 11 Reflections
- group communicated and worked well.

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Metal: Recieved





Cut extra metal after bend

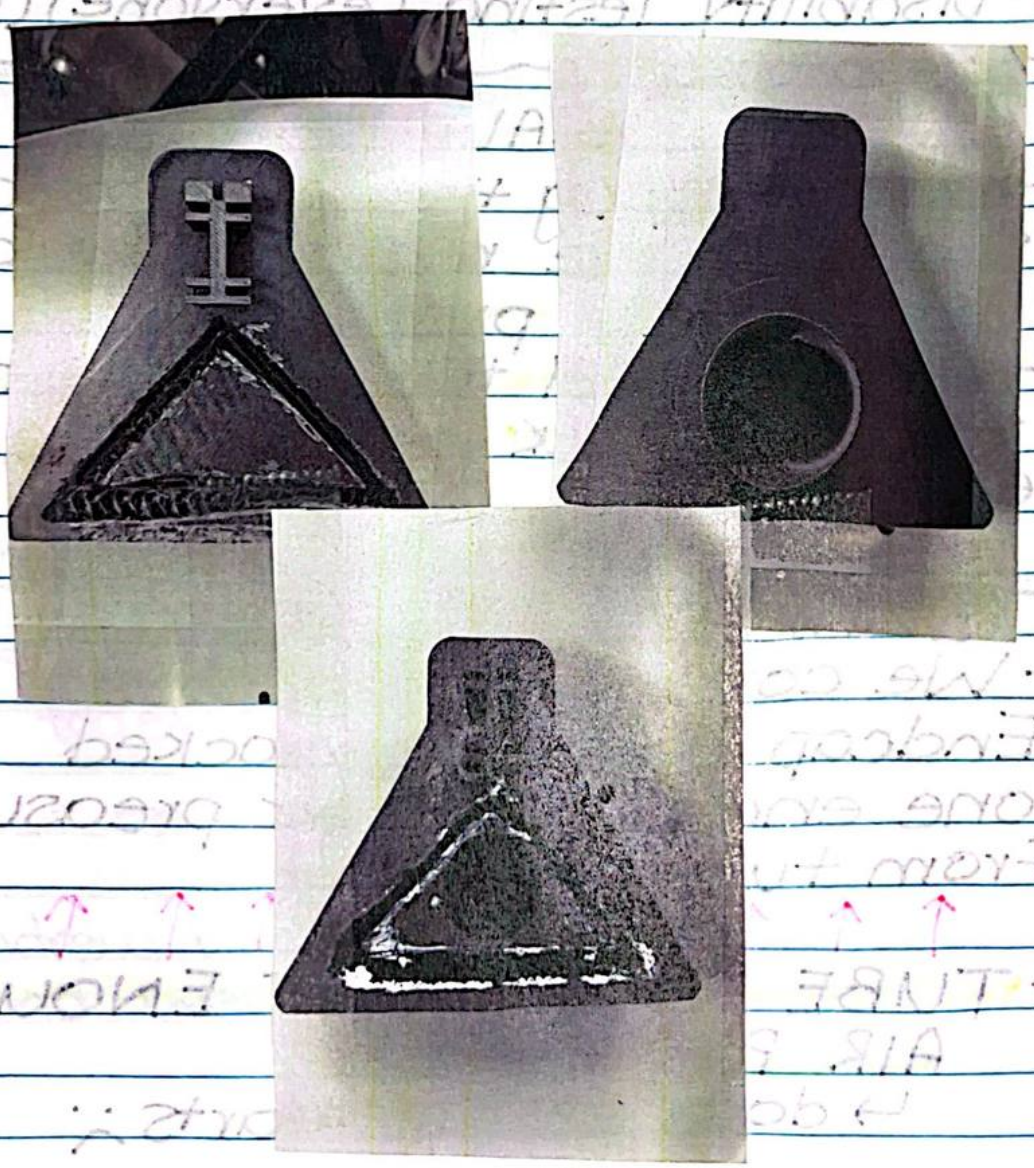
Andrew is cutting, bending, and welding track

Week 14

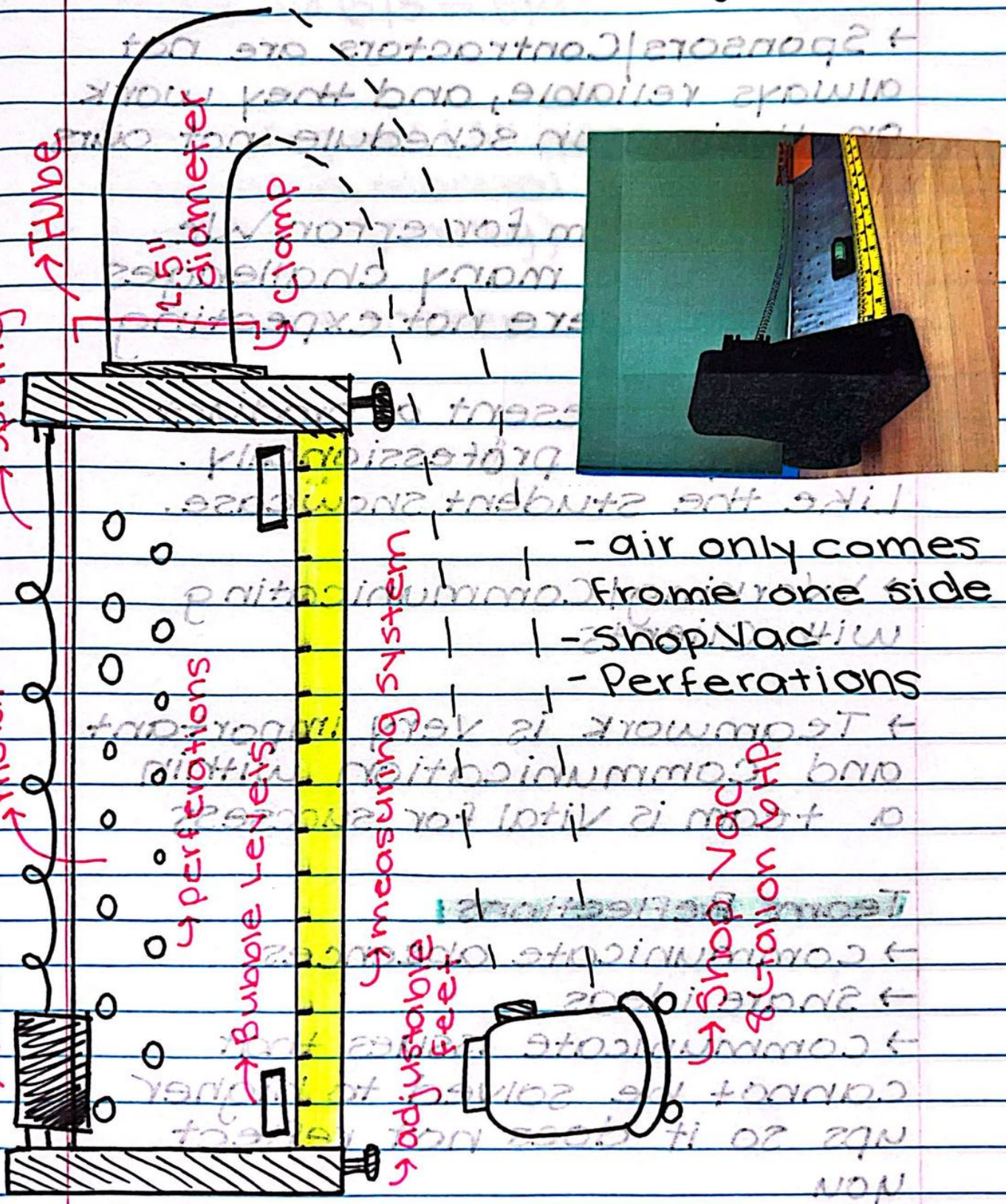
TESTING

- Shop-vac from shop lifts carts with air only coming from on end
- re-design one side Endcap to be closed: only one air entrance needed.
  - adapter no longer needed
  - Louis: edit CAD File

ENDCAPS



Final Design Drawing



- air only comes Fromie one side
- ShopVac
- Perferations

Reflections From Prodject

→ Sponsors/Contractors are not always reliable, and they work on their own schedule not ours.

→ Leave room for error. We ran into so many challenges that we were not expecting

→ How to present a product shortly but professionally. Like the student showcase.

→ Working/Communicating with Clients

→ Teamwork is very important and communication within a team is vital for succsess

Team Reflections

→ communicate absences

→ share ideas

→ communicate issues that cannot be solved to higher ups so it does not reflect you

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